

Proposal: Automated Corporate Outreach & Lead Generation System

1. Executive Summary

Currently, the Placement Cell operates on a high-effort model where Coordinators spend significant time manually scouring databases, extracting contact details one by one, and sending initial cold emails. This limits the bandwidth available for actual relationship building.

This initiative proposes an **Automated Outreach System** that handles the "top of the funnel" (prospecting and initial contact) autonomously. The system leverages the Apollo API, Generative AI, and automated logging to ensure Coordinators only step in when a high-value action (like a phone call or negotiation) is required.

2. The Shift: Current vs. Proposed Model

Feature	Current Manual Workflow	Proposed Automated Workflow
Data Sourcing	Manual search on LinkedIn/Apollo; copy-pasting details.	Automated extraction via Apollo API based on sector/hiring filters.
Outreach	Manual emailing or copy-pasting templates one by one.	Auto-mailing using verified IDs and standard templates.
Follow-up	Dependent on Coordinator remembering to check "Sent" folder.	Automated tracking ; system flags "No Response" leads.

3. System Workflow & Architecture

The system is designed to target **new** companies not currently in our fixed database to avoid overlap with existing relationships.

Step 0:

Using the Data available in the portal to make the list of active contacts which should not be approached by this system

Step 1: Automated Extraction (Apollo API)

The system queries the Apollo database for companies fitting our target profile/companies, It extracts the names and potential email patterns of HR decision-makers.

Step 2: Intelligent Email Guessing & Verification (Gemini + Verification API)

- **Pattern Matching:** The system uses **logic and internet** search (linkedin) using Gemini API to deduce the most likely email format (e.g., [firstname.lastname@company.com](#)).

Note: The Intelligent Email Guessing will be tested on previous data and the based on the number of results of wrong emails and right emails will decide whether to use this system or not

Step 3: Automated Outreach

- If the email is valid, the system automatically sends the standard Placement/Internship invitation email via the institute's official mail server with a coordinator's signature
- **Spam Protection:** Emails are "throttled" (sent in small batches, e.g., 50-60 per hour) to mimic human behavior.

Step 4: Centralized Logging (Google Sheets)

- All actions are logged in a Master Google Sheet accessible to the team.
- **Status Update:** The sheet updates in real-time: *Sent*, *Delivered*, or *Opened*.

Step 5: The "No-Response" Protocol

- If no reply is received within X days, the system triggers a "Level 2" action.
- It utilizes the Apollo API to **extract the mobile number** of the HR contact.
- This number is logged in the sheet with a flag: *"Action Required: Manual Call."*
- **Result:** Coordinators receive a curated list of people who need to be called, rather than searching for numbers themselves.

If the response is received it is mentioned in the google sheets and the respective coordinator will carry on further

4. Cost Analysis & Resource Utilization

The core cost driver is the **Apollo API**, which is highly cost-effective for the volume we intend to process.

Pricing Model (Apollo):

Cost Item	Short Term (Monthly)	Long Term (Annual Contract)
Base Cost	~\$109 USD (₹10,300)	~\$116 USD (₹11,000)
Included Credits	5,000	6,667
Eff. Cost/Credit	~₹2.06	~₹1.65

Credit Consumption Rates:

- **Email Extraction:** 1 Credit per verified email.
- **Phone Number Extraction:** 8 Credits per mobile number.

Efficiency Strategy:

We conserve budget by using a "waterfall" approach. We spend **1 credit** to email first. We only spend the expensive **8 credits** (for the phone number) *if* the email gets no response. This ensures we don't waste budget on companies that reply immediately via email.

5. Technical Stack

Component	Technology	Role
Core Engine	Python 3.11+	The "brains" of the operation. Handles API orchestration, logic flow, and error handling.
Data Sourcing	Apollo API	Primary source for corporate contacts and funding data.
Intelligence	Google Gemini Pro API	Used for context-aware filtering and inferring email patterns from unstructured web data.

Database (State)	SQLite / JSON	<i>Local Backend:</i> Stores temporary states (to prevent duplicate API calls if the script crashes).
Frontend/UI	Google Sheets API	<i>Coordinator Interface:</i> Read/Write access for team transparency. Acts as the "Database" for non-technical users.
Email Infrastructure	Gmail API (OAuth 2.0)	More secure than standard SMTP. Allows for tracking send status and prevents storing plain-text passwords.
Hosting (Prod)	AWS Lambda / Local VM	The script will run as a scheduled Cron Job (e.g., daily at 10:00 AM) on a secure institute server or cloud instance.

5. Implementation Roadmap: Testing to Production

To mitigate risk, we will follow a strict three-phase rollout strategy.

Phase 1: The "Sandbox"

- **Objective:** Validate the "Email Guessing" logic and API connectivity without sending external emails.
- **Activity:**
 - **Back-Testing:** Feed the system 1/2 *known* company mail and then guessing the new HR's email ID
 - **Metric:** Compare the system's "Guessed Email" vs. the "Actual Email."
 - **Success Criteria:** The system must achieve good accuracy in guessing email patterns before we proceed.
 - **Output:** A "Dry Run" Google Sheet populated with data, but with the email sending function disabled.

Phase 2: The "Canary" Deployment (Weeks 3-4)

- **Objective:** Test deliverability and domain reputation on a live but limited audience.
- **Activity:**
 - Target a low-risk sector (e.g., I4 companies)
 - **Volume Cap:** 100 emails/day.

- **Monitoring:** Manually check the "Sent" folder and "Spam" scores.
- **Coordinator Loop:** One designated coordinator will manually verify the "Call List" (Level 2 Action) to ensure mobile numbers are accurate.

Phase 3: Full Production

- **Objective:** Full-scale automated lead generation.
- **Activity:**
 - Remove volume caps (scale to **~500 emails/week**).
 - Enable full automation of the "No Response" protocol.
 - **Maintenance:** Weekly review of the Google Sheet to archive closed leads and monitor API credit usage.

6. Risk/Limitation Assessment & Mitigation

Risk 1: Domain Reputation (Spam Flagging)

- **Concern:** Sending bulk emails can trigger spam filters, harming the reputation of our .edu domain.
- **Mitigation:**
 1. **Strict Verification:** Email guessing is thoroughly tested on previous data and using 3rd party email verifying softwares
 2. **Throttling:** Limits are set to ~500 emails/day, spaced out over 24 hours.
 3. **High Reputation:** We leverage the historical trust score of our institute domain

Risk 2: Platform Dependency & API Volatility (Technical)

- **Concern:** The system is heavily dependent on third-party ecosystems (Apollo and Google). If Apollo changes their API pricing, rate limits, or data structure (breaking changes), or if Google updates their OAuth security policies, the automation could break overnight.
- **Mitigation:**
 - **Modular Architecture:** The code is written in isolated modules. If Apollo breaks, we can swap the "Data Fetcher" module for a competitor (e.g., Crunchbase or Lusha) without rewriting the whole system.
 - **Error Handling:** The script includes "Try/Except" blocks that alert the admin via WhatsApp/Slack immediately if an API call fails, preventing silent failures.

Risk 3: Cost Escalation vs. Low Conversion (The "Burn Rate" Risk)

- **Concern:** The system is designed to "escalate" to a mobile number search (which costs **8 credits**) if an email gets no response. If the market is unresponsive, the system might blindly burn through thousands of expensive credits on "dead leads" without generating a single interview.
- **Mitigation:**
 - **Tiered Spending:** Mobile numbers are only extracted for companies with >50 employees (High Value). Small startups are restricted to email-only contact

through Apollo to conserve funds and phone number of HR/founder of these companies can be taken out using tracxn

Risk 4: Technical Malfunction (The "Runaway Script" Risk)

- **Concern:** A slight logic error in the code (e.g., an accidental infinite loop or a failure to record "success") could cause the system to query the Apollo API thousands of times in a few minutes. This would drain our entire monthly budget (5,000 credits) in a single morning.
- **Mitigation:**
 - **Hard Budget Cap:** We will set a "Daily Spend Limit" directly in the Apollo dashboard settings. Even if the script goes rogue, Apollo will block requests once 200 credits are consumed in 24 hours.
 - **State-Aware Execution:** The Python script uses a local database (`state.json`) to "remember" which companies were processed. If the script crashes and restarts, it resumes exactly where it left off rather than re-processing (and re-paying for) the same companies.

Operational Limitations (Hard Caps)

These are the non-negotiable boundaries of the system:

1. **Volume Cap (Throttling):** The system is strictly hard-coded (gmail O-Auth) to send a maximum of **500 emails per day**. This is not a software limit but a safety protocol to protect our domain reputation.
2. **API Rate Limits:** The Apollo API allows a maximum of 100 requests per minute. The script includes "sleep" timers to ensure we never hit this ceiling and trigger a lockout.
3. **Data Freshness:** The system relies on the Apollo database. While generally accurate, there is a limitation where ~10-15% of data may be outdated (e.g., an HR director who changed jobs last month but hasn't updated LinkedIn).

7. Conclusion

By implementing this system, we effectively "clone" the productivity of our team. We move from a manual, linear process to an exponential one, where the software ensures that every potential recruiter is contacted, followed up with, and logged.